

Tailored Immersion: Adaptive Virtual Reality as a Tool for Personalized PTSD Treatment

The Role of Virtual Reality in Treating Post-Traumatic Stress Disorder

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This thesis explored the use of Virtual Reality (VR) as a tool for treating Post-Traumatic Stress Disorder (PTSD). Combining a literature review and semi-structured interviews with experts and individuals with PTSD, the study investigated how VR environments can enhance traditional therapeutic methods. The research focused on the potential of immersive and adaptable virtual environments to create controlled scenarios that adapt to individual needs. Key findings highlighted that VR-based therapy can provide patients with a safer, more engaging medium for processing traumatic memories by incorporating realistic, symbolic and multi-sensory stimuli. A prototype VR therapy system was developed as a proof of concept, illustrating how therapists could manipulate virtual environments in real-time to optimize therapeutic outcomes. This study offers insights into the role of user experience and technology design in developing therapeutic VR tools, with promising implications for improving mental health treatments.

CCS CONCEPTS • Human-Computer Interaction • Virtual Reality Treatment • PTSD Design

Additional Keywords and Phrases: Virtual Reality, Therapy, PTSD, Virtual Environment, Design

ACM Reference Format:

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1. INTRODUCTION

The thesis will focus on the role of VR (Virtual Reality) in the treatment of Post-Traumatic Stress Disorder (PTSD). The methods used were a literature review and semi-structured interviews. The key findings of these two methods will then be presented in a digestible format. The findings will then be attempted to be translated into design elements, relevant for the development of a VR-based therapy prototype for PTSD treatment. This thesis explores areas that are relevant

to Human-Computer Interaction (HCI), such as designing and implementing VR environments for PTSD treatment. The research connects with HCI through its focus on user experience within immersive environments and the need for personalization in VR therapy. By investigating how these elements can enhance therapeutic outcomes, this study contributes to broader discussions around user-centered design and the application of technology to improve quality of life. Though not building directly on specific HCI theories, the work highlights the importance of considering user needs and experiences in the development of therapy tools. Methods for therapy in this field are of great interest, not least due to the currently ongoing military conflicts in the world – but also due to an awareness that PTSD is a condition that affects not just people exposed to combat but traumatic events in general. Furthermore, research has observed that the efficacy of traditional treatments varies, and that patients may continue to have symptoms also after receiving treatment [19]. It is suggested that *poor emotional engagement* and *avoidance* as a coping strategy throughout therapy may be a reason [19]. This may explain the increased study of the use of VR in therapy, for treating not only PTSD but also of phobias and anxiety disorders [31]. Therefore, it is of great importance to examine the virtual environments in which the patient and sometimes the therapist are placed during this kind of therapy - as it plays a crucial role in future mental health interventions.

2. BACKGROUND

As a background, a short description about Post-Traumatic Stress Disorder as well as Virtual Reality and its variations will be given.

2.1.1. Post-Traumatic Stress Disorder

Post-traumatic stress disorder (PTSD) is a mental disorder classified as related to severe trauma- and stress experiences according to the Diagnostic and Statistical Manual, 5th edition (DSM-5) [2]. The diagnostic criteria for PTSD according to the DSM-5 is *the exposure to actual or threatened death, serious injury, or sexual violation* [2,32]. Examples of events may include war, natural disasters, sexual assault or domestic and other kinds of violence [10,15,19,33]. However, a diagnosis requires a number of specified symptom criteria to be fulfilled [3]. Four clusters of symptoms are used when diagnosing a patient with PTSD:

- Intrusion,
- Active avoidance,
- Negative alterations in cognition and mood,
- Marked alterations in arousal and reactivity [53].

It has been hypothesized that the various types of traumas carry different risks for developing PTSD, with the highest risk being associated with interpersonal violence [40]. The effects of PTSD on individuals vary, but the occurrence of intrusive thoughts or memories, which in turn evoke strong physiological and psychological reactions are common [69].

These processes of re-experiencing the trauma through memories, flashbacks or nightmares are characteristic of PTSD [2]. These may also be triggered by “safe” stimuli, and the person will therefore avoid the “safe” contexts that relate to these stimuli [69]. Additionally, delusional memories are major causes of mental anguish [73]. Other symptoms or attributes of PTSD include:

- Violent behaviors,
- Hyper-vigilance,
- Anger-management issues,
- Emotional dysregulation,
- Depression, and
- Anxiety [2,3]

2.1.2. Virtual and Augmented Reality

Virtual Reality (VR) uses technology to emulate an artificial environment where a person can interact with this environment using different kinds of controls. Through the addition of simulation technology, the user’s experience can be augmented with responsive stimulation of vision, hearing and touch [24]. This 360-degree interaction not only provides an immersive and three-dimensional environment but also allows users to interact freely with the virtual world [24]. Today the systems mostly employ a combination of visual and auditory perception, together with integrated control devices, either in the shape of handles or gloves that can perceive fairly detailed movements. A well-known example of VR equipment is the Oculus Rift [50], but there are many other competitors on the market. The wide range of applications that Virtual Reality (VR) and Augmented Reality (AR) offer can be aimed at healthcare, education, training, and as discussed in this thesis, therapy [13,30]. The usage of VR and AR with the aim to teach individuals complex medical teachings, such as surgery and anatomy, has been done with positive results [5].

2.2. RESEARCH QUESTION

The following research questions are posed as a base for the writing of this thesis:

- 1) What is the current status of the use of Virtual Reality and similar techniques in current therapy for Post-Traumatic Stress Disorder?
- 2) How can the present understanding of Post-Traumatic Stress Disorder and Virtual Reality treatments inform the design of VR-based PTSD treatment systems?

2.3. DELIMITATIONS

This study faced several limitations, particularly with regard to testing the prototype with users, which was outside the scope of this thesis. While it would have provided valuable insights, user testing was not feasible. The context in which

PTSD develops can vary, which reduces the generalizability of the work. Additionally, including more interviewees, particularly individuals with direct experience in VR therapy, could have further validated the prototype. The prototype has been developed with the primary goal of showcasing the functionality of the tool. Therefore, the quality of the graphics and visuals has been of secondary importance. The final design of the graphics will be different from the examples in the thesis.

3. THEORY AND RELATED WORK

This chapter will go through some of the most common methods used to treat PTSD conditions, without and with the support of Virtual Reality.

3.1. TRADITIONAL THERAPY METHODS FOR POST-TRAUMATIC STRESS DISORDER

Research supports trauma-focused treatments for PTSD. Key methods include:

Cognitive Behaviour Therapy (CBT): involves presenting traumatic stimuli in a controlled way, helping patients replace negative thoughts with positive ones. Components of CBT include exposure, cognitive restructuring, and anxiety management [9,56].

Cognitive Processing Therapy (CPT): focuses on identifying and challenging distorted thoughts related to trauma. Patients write down the causes and consequences of the trauma, which are then examined with a therapist [47].

Eye-Movement Desensitization and Reprocessing Therapy (EMDR): involves recalling traumatic memories while following a visual stimulus, such as a therapist's finger, which helps reduce the automated response to the trauma [49].

Exposure Therapy (ET): is a type of CBT which helps patients confront traumatic stimuli through gradual exposure, whether real or imagined. It is often used in conjunction with other methods like CBT and EMDR [19,54].

3.2. VIRTUAL REALITY FOR PTSD TREATMENT

In both research and clinical contexts, it is often difficult to recreate trauma-related scenarios effectively [12]. VR can help overcome this challenge by creating controlled virtual environments (VEs) that are adaptable to individual needs [41,42]. Examining a patient's behavior within a virtual environment can lead to insights into their reactions and knowledge [39].

3.2.1. VR-BASED THERAPY METHODS

Virtual Environments (VEs) are used in various treatments [19,31,42,65]. They are adapted to match the trauma of the patient [41,43] and it is common that the exposure is gradual [39,54]. Through virtual reality, it is possible to recreate an environment effective for a patient when an in vivo one would not be [7]. When immersed in a VE, a sense of presence can occur for the patient [4]. Sense of presence refers to a psychological state or subjective view where an

individual experiences a VE as if technology was not mediating the experience [4]. Immersion through stimuli lead to an increased sense of presence [58]. In other words, presence refers to the sense of being in the VE [68]. Patients are aware of that the environment is virtual and not real [23], but the effects of the experience in the treatment affect real-world situations positively [23]. PTSD may lead to avoidance of traumatic stimuli due to the associated negative responses it elicits [21,25]. With VR, however, a suitable virtual environment (VE) can be created beforehand, providing a continuous visual experience [33]. The immersive factor that VR provides is vital when treating PTSD [33], possibly because immersion leads to increased emotional engagement [19]. One example of ET in a VR setting is Virtual Reality Exposure Therapy (VRET), which is comparable to other types of psychotherapy [19]. Data suggests that VRET can be useful for patients who are resistant to traditional treatments [26]. There is also research on the expanded use of VR, which aims at extending the experience with added olfactory experiences [34] and other non-visual elements such as haptic feedback and medical enhancements, showing positive results [36].

Virtual Reality Cognitive Behaviour Therapy (VRCBT): involves using VR scenarios where patients practice mindfulness and other CBT techniques [64]. The aim is to improve the efficacy of CBT, by the creation of immersive environments [64]. Sessions are alternated with face-to-face sessions where beliefs, thoughts and issues are discussed [64].

Multimodular Motion-Assisted Memory Desensitization and Reconsolidation (3MDR): combines exposure therapy, EMDR and supportive counseling, often involving symbolic stimuli like images and music [37]. Patients engage in cognitive tasks during the sessions to desensitize traumatic [37].

Virtual Reality Exposure Therapy (VRET): provides controlled exposure to traumatic events in a VR setting [19]. Patients view these environments through head-mounted devices or large screens [19,25]. The stimuli can range from realistic to symbolic, and are tailored to each patient's trauma [19,25]. Studies suggest that VRET can reduce PTSD symptoms as well as associated conditions like anxiety and depression [44].

4. METHOD

This study's methodology will be qualitative, consisting of a literature review and semi-structured interviews, both of which will be analyzed via a thematic analysis. Furthermore, a prototype will be developed with the aim to showcase the current state of VR-based therapy for PTSD.

4.1. QUALITATIVE METHODS

Initially, a literature review was conducted; selection criteria were created, databases were chosen, search words determined. Semi-structured interviews were used to get a closer understanding of PTSD and its treatment. The interviewees in this study included experts that professionally treat PTSD, and, individuals who have PTSD, recruited via Discord. In total, three interviewees took part in the thesis, one professional and two individuals with PTSD. An additional expert validated the prototype but was not part of the thematic analysis. Individuals with PTSD may provide

information regarding their experiences or expectations of treatment and the possible medium of VR it would be offered through. Experts can provide theoretical and practical experience. The interview questions were created with the themes from the literature review in mind. The questions were open-ended and experience-based to encourage dialogue and ensure that the interviewees can express themselves. However, the questions for individuals with PTSD were adapted to be sensitive and ethically considerate. The interviews were conducted through Zoom, as physical interviews were not feasible. Before the interview, the interviewee was provided with a general overview of the study, informed that they can withdraw without cause, and assured that their data will be anonymized. The anonymization of the interviewees with PTSD was done thoroughly, as these individuals are viewed as vulnerable. The general principles of research ethics were followed and applied, regarding the collection, examination, analysis and presentation of data [72]. The literature review, before it was conducted, had several key areas that needed to be examined and determined.

The inclusion criteria can be viewed in appendix B, table 1. As shown in appendix B, table 2, only one set of keywords was used, as it yielded sufficient results focused on VR, PTSD and VEs. Of 153 examined articles, 20 were included in the review, with others excluded for not addressing the research questions or lacking depth. The selected articles, listed in appendix A, were analyzed thematically, with the results presented in the analysis chapter. The qualitative findings gathered from the literature review and interviews were analyzed using thematic analysis, a widely used method in qualitative research [45], and is known to be a flexible method [8]. The analysis involved extracting relevant information, generating codes, categories and developing themes. Codes were reviewed for patterns and grouped into categories, and later themes [8,67]. The semi-structured interviews allowed for flexibility in questioning [16], ensuring that themes varied between experts and users. Finally, a prototype was developed to explore and evaluate the practical application of the concepts discussed. The concepts were based on the results of the thematic analysis of a literature review and the interviews.

5. ANALYSIS

The aim of this chapter is to present the results of the thematic analysis of the literature review and the interviews.

5.1. LITERATURE REVIEW - THEMES

Through thematic analysis, eight themes have been identified. The table below provides an overview of these themes, while a comprehensive account of it can be found in the appendix C for those interested in further exploration.

Table 1: Author, title of the themes and their relation to the author.

Author	Multifaceted Role of Virtual Environment	The Impact and Essence of Immersion	The optimization of Virtual Environment for Effective Treatment Outcomes	Auditory Stimuli	Tactile Stimuli	Visual Stimuli	Olfactory Stimuli	Move ment
[41]	X	X	X	X	X	X		X
[20]	X	X	X	X	X		X	
[61]			X					
[46]	X	X						
[25]	X	X	X	X		X		X
[77]	X							X
[76]	X	X	X					
[51]	X	X	X	X	X	X	X	X
[70]	X		X			X		
[18]	X	X	X					
[55]	X	X	X					
[57]	X	X	X	X		X	X	X
[44]	X	X	X					
[22]	X	X	X				X	
[28]	X	X	X	X				
[48]	X	X	X					X
[63]	X							X
[38]	X	X	X	X	X	X	X	X
[75]	X		X					

[71]	X	X						
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5.1. THEMATIC ANALYSIS – SEMI-STRUCTURED INTERVIEWS – EXPERT INTERVIEWEE (E1)

Through the thematic analysis, six themes have been identified. The table below provides an overview of these themes, while a comprehensive account of it can be found in the appendix C for those interested in further exploration.

Table 2: Title of the themes and their relation to the interviewees.

Themes	P1	P2
Adapting Therapeutic Approaches to Individual Needs	X	
Creating Safe Spaces in Therapy	X	
Patient-to-VR Specific Considerations	X	
Challenges with Traditional Therapy		X
Perceptions of VR Therapy		X
Features of VR Therapy		X

5.2. THEMATIC ANALYSIS – SEMI-STRUCTURED INTERVIEWS – EXPERT INTERVIEWEE (E1)

Through the thematic analysis, six themes have been identified. The table below provides an overview of these themes, while a comprehensive account of it can be found in the appendix C for those interested in further exploration.

Table 3: Title of the themes derived from the expert interviewee.

Themes
Innovative Therapy through VR
Patient Experience and Safety
Personalization and Adaptive Therapy
Communication and Emotional Processing in a Virtual Environment

Navigating Ethical Terrain in VR Therapy
Cognitive and Behavioral Change via VR

5.3. RESULTS FROM THE ANALYSIS

This chapter presents the results of the thematic analysis conducted on literature and interviews. 17 themes emerged in total: eight from the literature and 12 from the interviews. Information from the literature is referenced accordingly, while interviewees are denoted as Person 1 (P1), Person 2 (P2), or Expert 1 (E1). Statements related to the research questions are indicated as Research Question 1 (RQ1) and Research Question 2 (RQ2).

5.3.1. LITERATURE AND INTERVIEWS – RELATION TO RESEARCH QUESTIONS

The literature review, interviews and thematic analysis revealed various roles VR can play in therapy. Virtual Environments (VEs) are highlighted as powerful tools in trauma therapy due to their ability to provide personalized and adaptable experiences for patients [18,41,77]. Therefore, a key theme throughout is the necessity for personalization and adaptability in VEs [18,25,41,44,51,76,77]. The ability to adjust in real-time and utilize multi-sensory stimuli, as to tailor to the patient's individual needs [25,38,41,51]. *This relates to RQ2 by highlighting the understanding of PTSD and VR treatments can lead to design of more effective VR-based PTSD treatment systems. More specifically, the ability to adapt and tailor VEs to address individual patient needs is a crucial consideration when designing such systems.*

Interviewees discussed negative experiences with impersonal therapy [P1, P2] and emphasized the need for adaptation to individual needs [E1]. Balancing efficacy and safety in VR are crucial [44,55,70,76]. *This necessity highlights the current state of systems, where real-time monitoring might be rare or absent, but would have an impact on both mentioned factors (RQ1). It also highlights the need for safety mechanisms within VR systems, such as monitoring but the ability to adapt to ensure patient comfort and safety (RQ2).*

VR's innovative capability broadens the variety of effective triggers for patients [E1], but also raises safety concerns, necessitating risk management [E1]. Monitoring patients' reactions before exposure is essential [P1, E1]. Authors noted VR's potential for realism and patient engagement [25,44,57,70] which is important for effective treatment. Patients also stressed the importance of safety and comfort [P1, P2], suggesting methods like bringing a support animal for reassurance [P1]. The use of symbolic interaction is an example of an alternative method for direct exposure, as it has been shown to be just as effective as replicating the trauma itself [20,25,46]. Symbolic and indirect exposure, such as using a flying carpet to simulate driving, can provide safer alternatives to direct exposure [18,20,25,41,46].

The discussion of symbolic interaction and indirect exposure relates to RQ2, that VR design could include flexible options in regards to how the patient is exposed to traumatic stimuli. This consideration could aid in achieving therapeutic goals, since replicating the trauma can be too intense for some patients and other alternatives are needed. This aspect

is especially relevant for patients with conditions like aphantasia [P1], who cannot visualize mental images and thus benefit uniquely from VR's visual and immersive capabilities. None of the interviewees had direct experience with VR-based therapy but agreed that it has powerful potential [P1, P2, E1], *which supports the relevance of RQ1. This is due to the agreed potential of VR, and underscoring the importance of integrating current therapeutic knowledge into VR design.*

"... but I often have thought back to moments that have been really hard and like, stained me. I wish I did something different back then. And I feel like VR really gives me an option to do that, to do it differently." – P2

The identified impact of VR can be attributed partly to the effect of immersion it can provide, as it can induce the feeling of existing within the virtual space, evoke emotional reactions, and allow for a realistic but engaging user experience [25,41,46,57]. Realism and immersion enhance emotional and cognitive engagement, improving therapeutic outcomes [E1]. Immersion can be achieved through sensory, spatial, emotional, and cognitive engagement [25,41,76]. The sensory aspect relates to various stimuli, such as auditory, visual, and tactile [41,51], though olfactory remains less commonly used. Tactile stimuli have been used to re-enact realistic physical sensations, enhancing realism and patient engagement, which is vital for processing traumatic memories [38,71]. Movement, another vital stimulus, enhances immersion and therapeutic outcomes [25,41,46,57]. Customized therapy ensures content remains safe and effective for patients [E1]. Patients can practice real-world interactions in a safe setting, allowing for the exploration of different scenarios and emotional responses [18,38,76]. A critical component of therapy is patient expression, as effective communication is linked to therapist qualities and the patient's emotional processing ability [E1]. A negative aspect of VR is the risk of disorientation, discomfort and the feeling of isolation for the patient [48,55,63,70].

This relates to RQ1 by addressing the current limitations and challenges faced in VR therapy, which are important to identify when assessing the state of VR-based treatments. The qualities of the therapist have an impact on therapist-patient communication, which in turn affects the patient's ability to process emotions within the VE [E1]. Therapy also involves memory restructuring through behavioral changes [E1]. Treatment plans and evaluations are crucial, and therapist competence significantly affects treatment efficacy [P1, P2, E1, 43,46]. *This relates to RQ1, as this reflects the current status of VR therapy and emphasizes the importance of therapist skill in its success. Furthermore, these findings suggest that VR systems should not only be designed for the patient in mind, but also with tools to support the therapist in training (RQ2).* Co-existing in a virtual space requires diverse skills from therapists. Monitoring tools can gauge patient reactions and adjust exposure systematically [20,41,51,55,57,61,76]. And the level of adjustment to the exposure can be gauged by the reaction, which will be classified and ranked, for a more systematic approach. Which is not unheard of in traditional therapy, to establish and gauge reactions before exposure [E1], but without the monitoring equipment. *This reflects on both RQ1 and RQ2, suggesting that the current VR systems need tools for real-time monitoring and adaptation, which could be the focus of future VR-based PTSD treatment designs.* The broad

exposure capabilities of VR therapy raise ethical concerns. Interviewees highlighted the need for responsible VR use, possible supervision, and individualized therapy with thorough risk assessment [E1].

5.4. LITERATURE AND INTERVIEWS – FOUNDATION FOR VIRTUAL ENVIRONMENTS FOR VR-BASED THERAPY

The following findings can be translated into design elements for virtual environments, organized by sensory categories to reflect VR's integration capabilities. Appendix D contains interview excerpts with actionable information for design, combined with literature findings in the subchapters below.

5.4.1. VISUAL – AVATAR, OBJECTS, AND THE ENVIRONMENT

Visual stimuli are the most dominant in a virtual environment, from the avatar representing the user's body, to the surrounding environment and the objects the user can see and interact with. *These elements are important in understanding how VR systems are currently used in therapy, as they directly influence the patient's experience and immersion within the therapeutic environment (RQ1). Furthermore, the design and implementation of these visual stimuli can inform the development by ensuring that the virtual world is both engaging and therapeutic (RQ2).* The suggestion of a tutorial for usage of VR, before the VR-therapy sessions are continuous, could be beneficial as to decrease the possible stress for the patient [P2, 43]. *This aligns with RQ1, as the literature suggests that easing patients into VR environments could improve their overall therapeutic experience. By including such tutorials, VR-based treatment systems can be designed to minimize patient stress and improve upon treatment outcomes (RQ2).* It was also mentioned that, there should be something that indicates for the patient when a session is active or not [P2]. All three interviewees touched upon the topic of it becoming too intense for the patient, and the suggestion of having the function to stop or exit the environment was discussed. However, the expert reasoned that a stop button, which the patient controls, would allow for further avoidance [E1]. Instead, the therapist should have the capability of stopping the session if needed, but also adjust the current exposure the patient is subjected to [P1, E1]. *This insight connects to the current use of VR systems in PTSD therapy (RQ1), lifting the importance of controlling the intensity of the experience within the VE to avoid overwhelming the patient. Furthermore, it also informs the design of VR systems by suggesting that therapists, rather than patients, should manage exposure levels and adjustments to ensure the effectiveness of the treatment (RQ2).* The ability of changing the virtual environment, such as weather, via an interface that the therapist has control over is not unheard of [41]. Furthermore, it would be possible, if needed, to include a support animal, as it could help the patient in recovery [P1]. *It points out how the current use of VR in PTSD therapy can be enhanced by allowing therapists to customize the environment (RQ1), ensuring that the therapy is tailored to the individual needs of the patient (RQ2).* The idea of a patient and therapist co-existing, each with their own avatar, within the virtual environment have been mentioned by the interviewees [P2, E1] and in the literature [25,44,55]. Furthermore, it could be quite powerful, as it could allow for the patient to remain in the environment, instead of removing the VR-headset during session. *This highlights how current VR systems are used in therapy (RQ1) and suggests that the co-presence of*

avatars could improve the therapeutic relationship and increase patient engagement, thereby informing the design of more effective VR-based PTSD treatments (RQ2).



Image 1. Example of gradual introduction of traumatic elements (Authors figure).

When subjecting the patient to triggering objects, it is suggested to do so in a gradual manner. For example, introducing the patient to the environment first, then the objects, one at a time, as the patient becomes more resilient [E1]. *The gradual introduction of traumatic elements reflects current practices in VR therapy for PTSD (RQ1) and point out the importance of VR systems that can accommodate a gradual approach to exposure therapy, tailored to the patient's resilience (RQ2). Furthermore, tailoring, to match the patient's resilience, real-time monitoring may be the approach to achieve this (RQ2).* Image 1 illustrates the gradual introduction of traumatic elements in therapy. It starts with the environment, adjusts the time and weather to match the memory, and introduces symbolic vehicles. Then, smoke is added, followed by military vehicles replacing the symbols. Finally, the scene is completed with smoke and fire emerging from the military vehicles. With objects, it relates to the specific representations relating to the patient's trauma, for it to be effective in triggering [E1]. *The connection to the patient's trauma, that VR systems are used to trigger responses in PTSD therapy (RQ1), highlights the need that the design of these systems need accurate and personalized representations of traumatic events (RQ2).* The virtual environment, with objects or another stimulus,

whose aim is to trigger the patient, must be designed individually [E1,20,27,43,46,54,81,82]. However, the manner of doing so offers more than one route, since moving away from realistic depictions of the patient's trauma still is proven to be effective in triggering [18,20,25,41,46]. *The ability to have flexibility of VR systems for PTSD therapy (RQ1) and the suggestion that both realistic and symbolic representations can be effective, which enriches the adaptability of VR system design as it can match different patient needs (RQ2).*

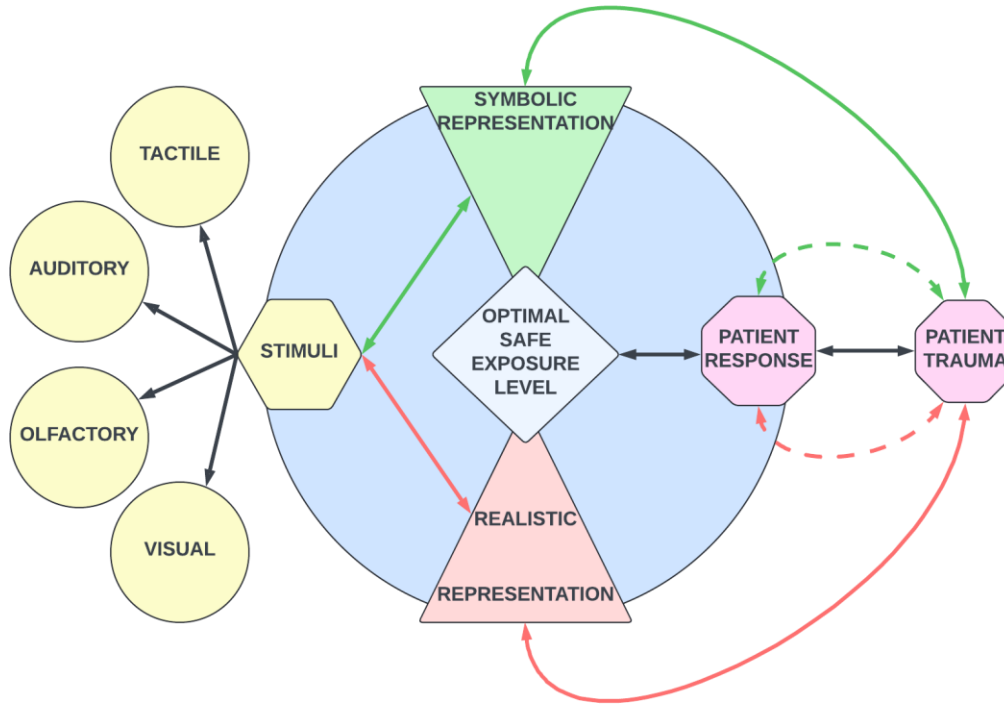


Image 2. Balance between Realism and Symbolism for Safe Exposure (Authors figure).

Image 2 illustrates a conceptual process for safely exposing patients to their trauma. The exposure can range from symbolic to realistic representation, depending on the patient's needs and responses. Different sensory stimuli are used to create an optimal exposure level, as each patient may react uniquely to either symbolic or realistic triggers. The creation of the environment should be adapted to match the patient's trauma[18,70], such as a forest or specific scenes like a car crash. Other factors related to the environment such as the weather, the behavior of other humans or the layout, can affect the response and perceived realism from the patient [18,70]. *These environmental factors are crucial in the current use of VR therapy (RQ1) and should be carefully considered in the design of VR systems to ensure they accurately reflect the patient's traumatic experiences (RQ2).*

5.4.2. AUDITORY AND TACTILE - IMMERSION, THERAPEUTIC BENEFITS AND ENVIRONMENT

Through the virtual environment, the user exists and may traverse through the environment. And as the aim is to have a sense of immersion, the utilization of auditory stimuli can enhance the experience. Sound can take many roles, such as to indicate for the patient that the therapeutic session has started or ended [P2]. *The various uses of sound are an important aspect, as it can act as cues within the therapeutic process as a guide (RQ2).* This is like what audio has been used for, as to indicate a warm-up but also the end of a session [25,41]. It can also improve the session, as it could enhance the immersion but also act as a trigger during therapy [E1]. However, the type of sound, and what would elicit it is important to consider, as to match the patients' preferences [25,41]. For example, the selection of images and music that are related to the traumatic event, such as a deployment to a warzone [25]. And in contrast, music from the present-day can act as a representation of current time for the patient, but also as an end of a session [25]. Furthermore, the literature proposes an interface which allows the therapist to add or remove stimuli, such as explosions or gunfire [41]. Such actions would be used later in the therapy, as to focus on specific portions of the trauma [41]. With multisensory, such as visual, audio, and tactile, VR systems can provide the user an improved sense of presence, and on their memory for the environment and related objects [51]. *The multisensory approach is already used in VR therapy for PTSD (RQ1), but it suggests that future VR systems should continue to integrate these sensory elements to enhance patient immersion and treatment effectiveness (RQ2).*

6. VR-BASED PTSD THERAPY PROTOTYPE

This chapter contains an exploration of a possible therapy process that relies on VR as a support tool.

6.1. IDEA DESCRIPTION.

The initial step the therapist would do is an assessment to understand the patients background, symptoms, history and specific triggers [74]. This assessment usually involves questions about their medical history, previous therapy experiences, current medication, and lifestyle factors. A common assessment tool, and sometimes used throughout the treatment, is the PTSD Symptom Scale, as a way to measure symptom severity [11]. During this phase, the patient shares their experiences and challenges, to help the therapist identify key areas to focus on during therapy, such as creating a hierarchy of relevant traumatic experiences [74]. This background collection helps ensure that the treatment addresses the patient's specific needs [75]. The patients would receive education about the VR-therapy process, how it works and what to expect [75]. The illustrations below follow this winter forest example, starting from a low-level 2D visualization, increase in detail, and lastly a VR version. The description for this scenario can be found in appendix E.



Image 3. *Simplistic 2D scene with stick figures (Authors figure).*



Image 4. *More detailed scene that includes elements relevant to the traumatic experience (Authors figure).*



Image 5. The VR version of the previous scene focuses on 3D and a more realistic environment (Authors figure).

6.2. EXAMPLE CASE PRESENTED VIA VR-PROTOTYPE

In this chapter, we will explore an example case of VR-therapy being applied for two fictional patients: one dealing with trauma related to sexual assault, and the other with war related symptoms. The case regarding sexual assault has been explored above, based on a fictional description. The war-related case is based on a description that can be found in appendix E, text 2. Furthermore, creating an exact replica of a scenario is not always necessary, as both realistic generic scenarios and abstract have been found to be effective [7,21]. In the attempt to adhere to the two case descriptions (text 1, text 2) which can be found in *appendix E*, while remaining grounded in the literature, a requirement specification was made (see appendix F). The development of the two scenarios that make up the VR-prototype started with the immediate environment where the user would find themselves in. The requirements that were implemented into each environment can be viewed in *appendix G*.

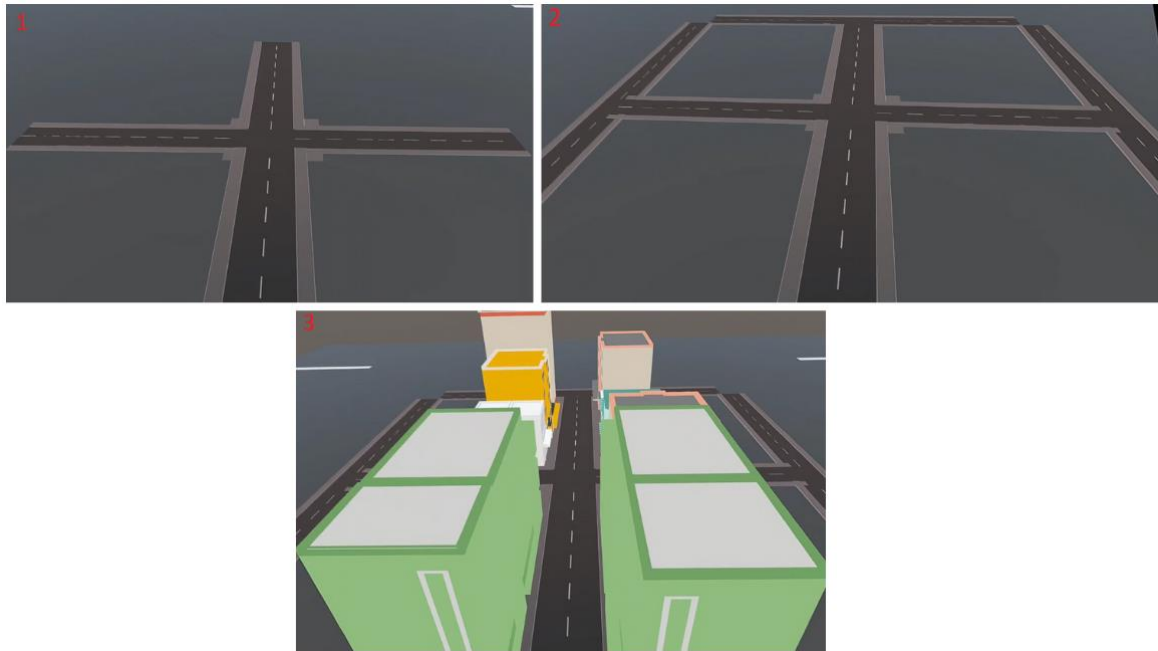


Image 6. The first three (out of ten) levels of progressively increasing stimuli regarding the immediate city environment (level 1-3) (Authors figure).

In image 6, what is visualized is the city environment. The first two images represent the roads and are the lowest possible state of visual representation possible within this prototype. The literature that this design choice was based on relies on the therapist's ability to alter the virtual environment [26,52]. Furthermore, the ability to decrease and increase the stimulus of environment has been noted [48,54,73]. The aim is to introduce the patient to an environment that gradually becomes more relevant to the actual traumatic experience. However, doing so in a way where the risk of being overwhelmed is to be mitigated, while still having a connection to the actual experience itself. For example, the road is an object found in the traumatic experience but is not expected to be the catalyst for the patient being avoidant. The first image represents the area where the patient would navigate, while the second is to prevent that the immersion breaks. For example, if the patient navigates themselves close to the end of a central road, there is a continuation of the landscape which may provide the feeling of a continuous existing city. In the last image, buildings emerge, which are placed to allow pedestrians to maneuver on the sidewalk. The buildings themselves have been selected for their height and size, but also the different appearances they have, as to induce a feeling of realism of being in a city.

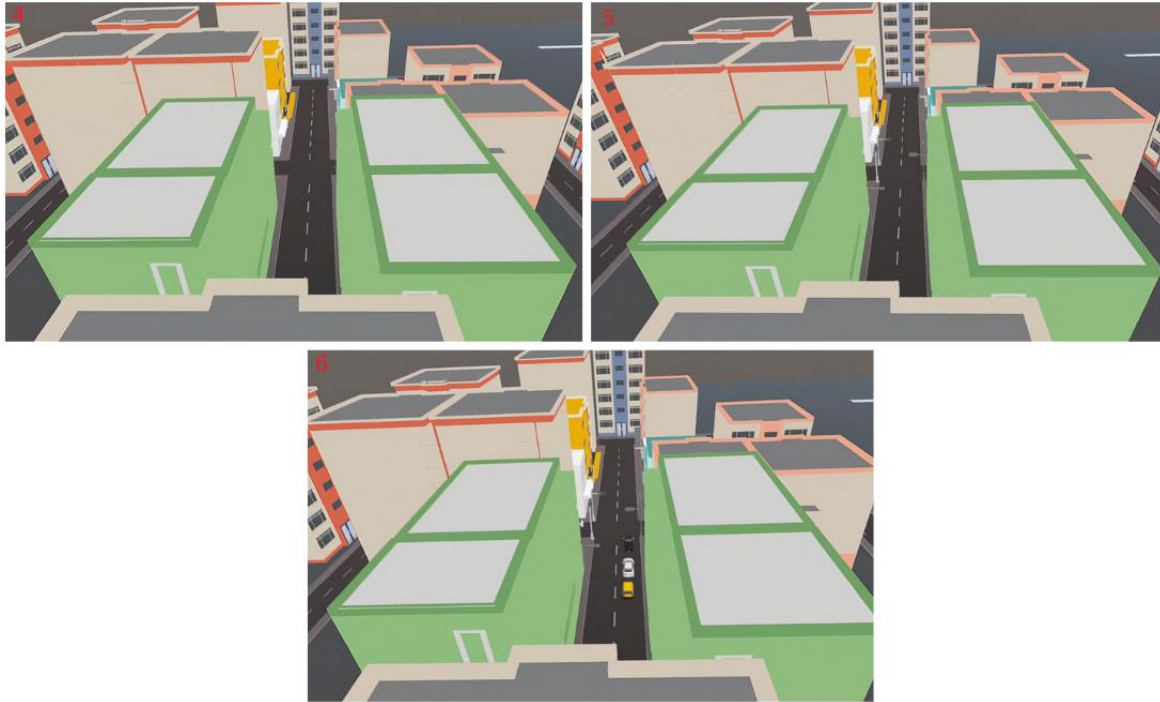


Image 7. The next three levels of progressively increasing stimuli, built upon the previous stages (level 4-6) (Authors figure).

In image 7, more buildings are added in the next (fourth) level of increased stimuli. The buildings are placed at the end of each road with the intent that if the user is in the middle of the intersection, and looks down each road, their view would eventually be obstructed by a building. The aim is to maintain a feeling of immersion, and that the city is larger than what it actually is. And to further improve the realism of the environment, the next level adds details such as streetlights, and traffic lights for vehicles. The level that follows adds civilian vehicles to the environment, placing them onto the roads, and that either they are waiting by a traffic light or that they are driving away from the intersection. The addition of the vehicles is primarily for the purpose of immersion, as the description of the experience includes military vehicles.

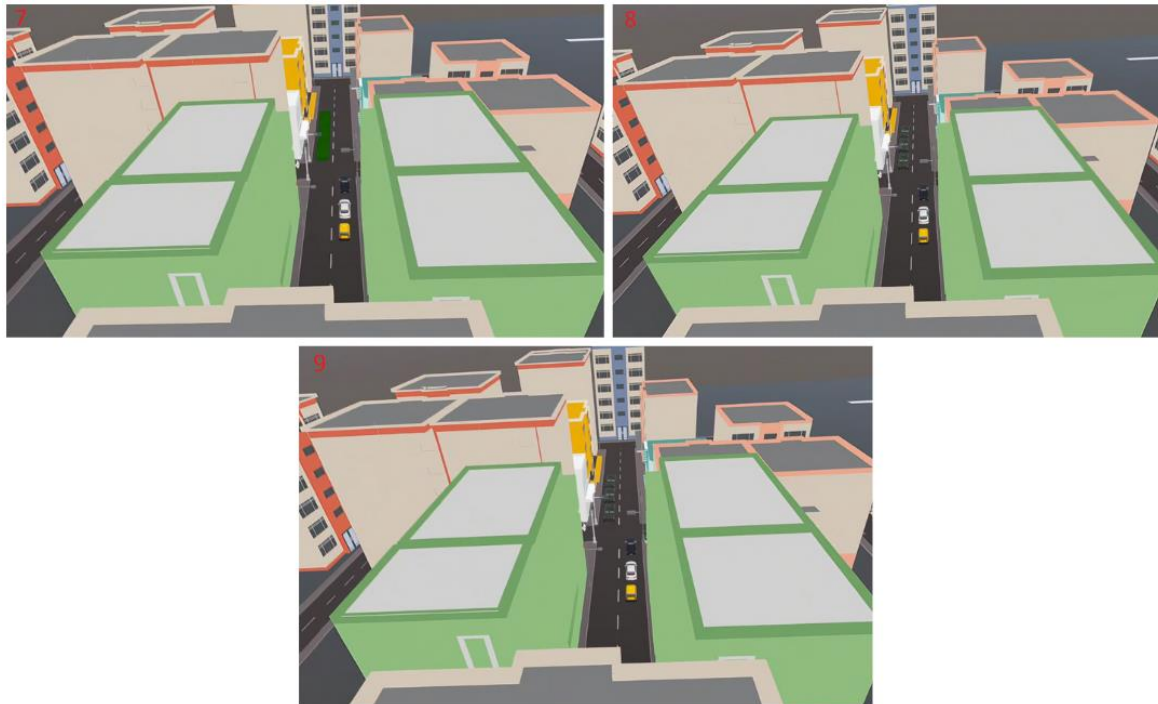


Image 8. The last four levels of progressively increasing stimuli in the city environment (Authors figure).

After the civilian vehicles have been added, the level that follows introduces the military vehicles. In relation to the real experience, the location of the vehicles mimic reality, but the visual representation is purposefully poor in detail except the color green. It was believed that the color green would represent the military, as it is quite common that military equipment has that color. More specifically, it is an attempt to lessen a potential negative reaction, since the visual stimuli would rely more on symbolism than realism due to the lack of graphical details (see image 13). The level that follows replaces the low-graphic military representation, with realistic military vehicles. After all the vehicles are present in the environment, the following level adds civilians to the environment. The reason why a realistic military vehicle was deemed to pose less of a risk, in terms of overstimulating the patient, was that vehicles are inanimate objects while people are not. It can be reasoned that the patient encounters people in their lives, without being triggered, and therefore a military vehicle could cause a stronger reaction. But on the other hand, the presence of civilians could indicate that the traumatic situation is more likely to happen, and if it did, the chaos that ensues would be found in the actions of the people, not the vehicles. Indeed, a burning vehicle at the front of the line is involved in the traumatic event, but the effect that gunshots and explosions have would be found in the people around. For example, in this prototype, when the triggering events occur, civilians would be animated and appear to be running away.

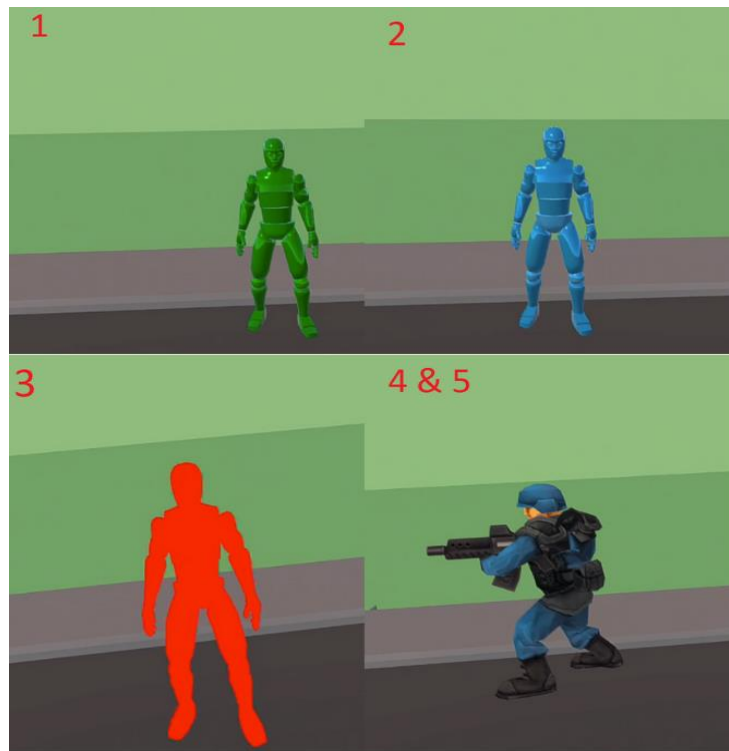


Image 9. Five states the aggressor figures have in the city environment (Authors figure).

In image 9, there are four different visualizations of the aggressor figures, but the last one counts for two states. In a final implementation, the granularity should probably be larger overall, with maybe six or seven different degrees of realism, to achieve an adaptable environment and a gradual increase [48,54,73]. Here the first three share the same shape, but with different colors. It was assumed that green would be perceived as less threatening, compared to the red figure. The fourth and fifth state of the aggressors are limited to the city environment, and the figures were chosen to illustrate the possibility of a more precise visualization of aggressors. Furthermore, all the states of the aggressors have animations related to their movement, that they have an idle or walking animation. However, the last state makes the soldier figure do a shooting animation, as if it was firing its rifle. The main reason for making two versions of the same figure was to allow for a gradual increase in exposure. More specifically, without the shooting animation of the last state, the figure could be perceived to be less of a threat.

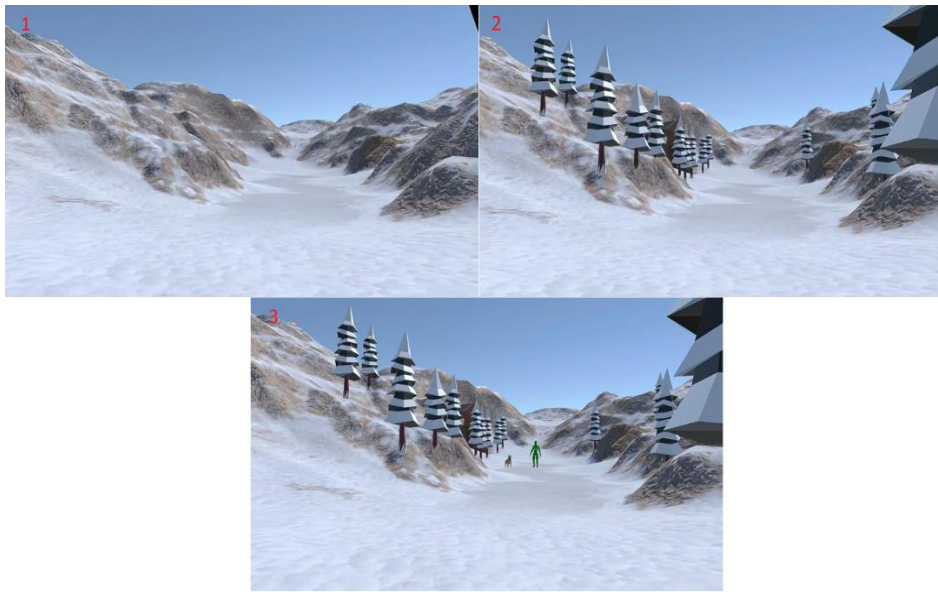


Image 10. *Three levels of progressively increasing stimuli of the forest environment (Authors figure).*

The three levels that represent from minimal to full environment are visualized in the above image. As the previous scenario, the attempt to connect with the traumatic event while maintaining minimal exposure is seen in the first level. There were considerations to attempt to lower the possible stimuli, but it was disregarded due to technical limitations and the risk of losing the connection to the event itself. The following level introduces the objects that fill the environment, trees and a cabin further in. And finally, the figures that represent the aggressors in the traumatic event, including the dog that was described.

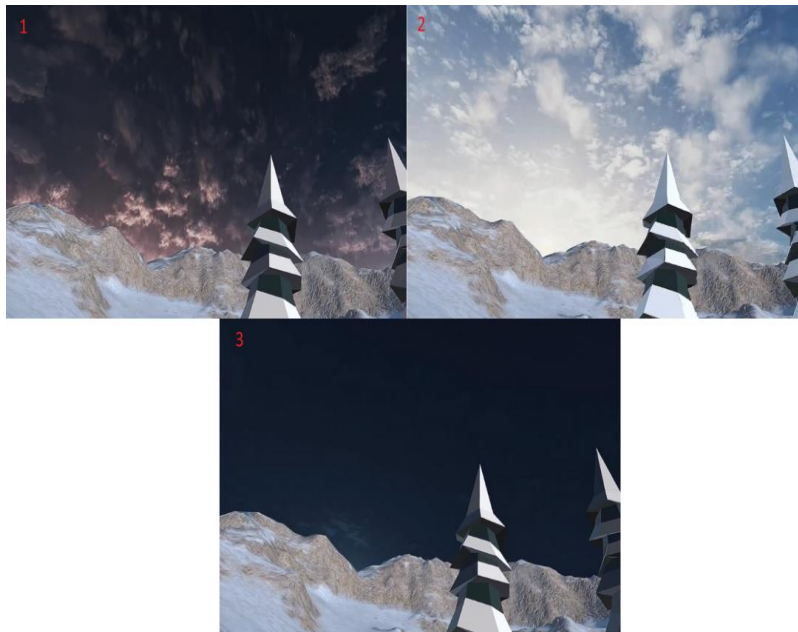


Image 11. *Three types of skies in the winter environment (Authors figure).*

Beside the immediate environment, aspects such as weather and time were explored, and implemented in both environments. Three unique skies were added and are visualized in the above image. The reasoning for why is to allow for a more realistic representation of the traumatic event, but also to alter the environment, as the exposure progresses. For example, a non-accurate sky could be used initially during the exposure, as it would stray away from the event that happened, but later be replaced with a more accurate one for improved realism. Beside the visualization of the sky, the ability to alter the time of day, and being able to adapt the environment to any time of the day was seen as an important implementation. Note that time adaptation affects the environment by visually representing changes from dawn to dusk, which includes the rising and setting of the sun.



Image 12. *Snowing effect used in combination with adapting the time of the day (Authors figure).*

Beside the ability to alter the sky and time of day, a snowing effect was implemented and is visualized in the above image. The aim was to showcase the possibility of introducing further elements that can have an impact on immersion and realism, thereby strengthening the therapeutic potential.

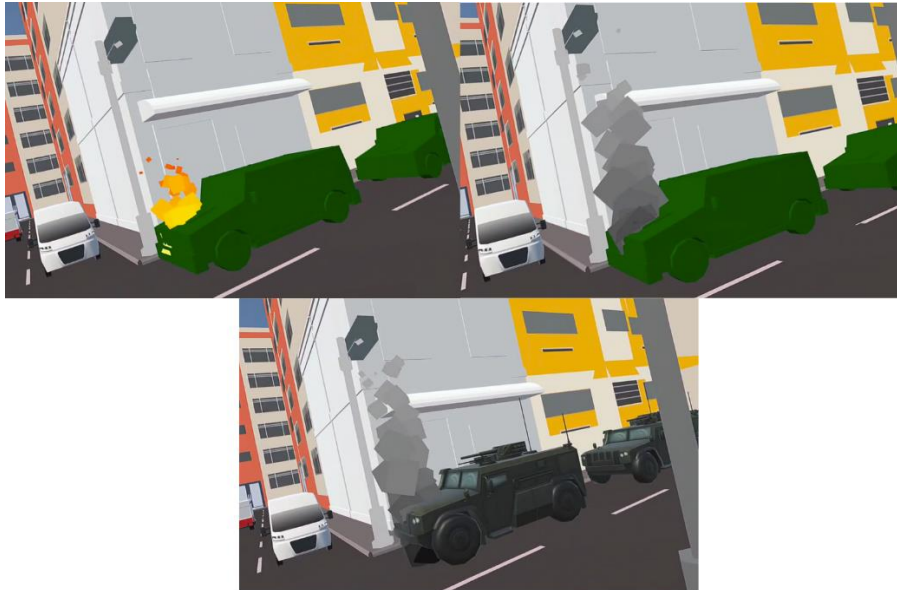


Image 13. Effects visualized with two different states of military vehicles (Authors figure).

Image 13, it illustrates how to visualize smoke and fire for the patient. Like the previously mentioned snow effect, smoke and fire were implemented to further enhance the effect of the environment. The choice was motivated by the description of the traumatic event and was therefore deemed important to implement to further complement the previous efforts of illustrating the environment.

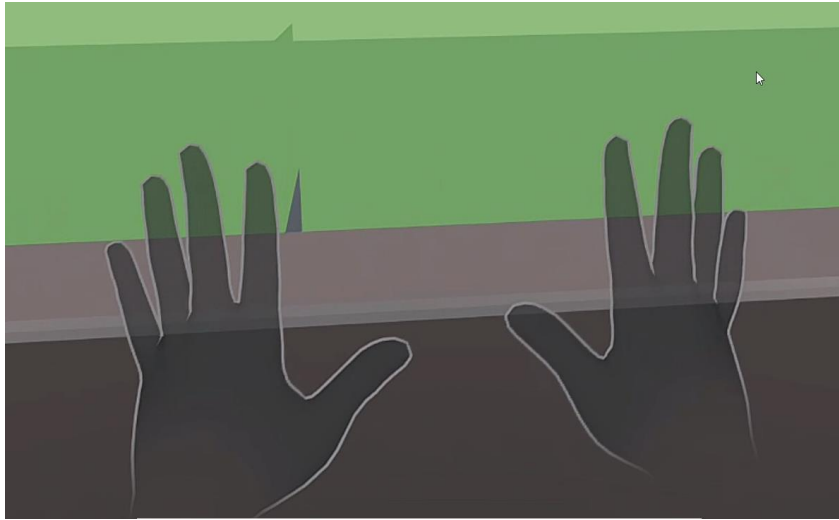


Image 14. Hands models which represent the user's hands (Authors figure).

In regard to immersion and agency, the visual representation of the patient's hands was implemented in both environments. It was deemed of high importance, as the literature points out the value of having a digital representation of the user's limbs [45,48,56]. Furthermore, the literature suggests to give patients agency within a virtual environment [36], as well as an ability to interact with it [44,54]. Therefore, the hands can interact with specific objects and are continuously updated to match the user's movements. Considerations to use models of the the hand-held controllers primarily used handling VR-systems were dismissed due to the negative effect it could have on immersion.

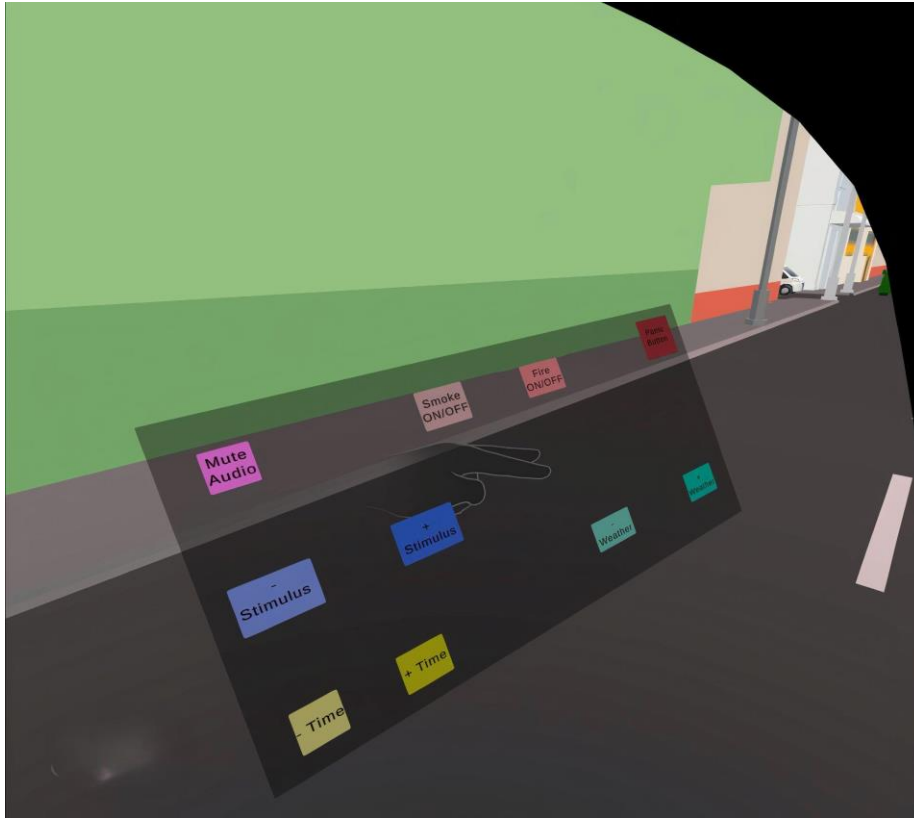


Image 15. Hand-menu presents all functions for the city environment (Authors figure)

Image 15 shows a menu (only available to the therapist). Both the literature and the interviewees lifted the importance of the therapist having control of the environment [P1, E1, 28,55]. Therefore, the goal of the hand-menu is to allow for quick alterations of the environment without removing oneself from it. As mentioned before, the ability to decrease and increase the levels of the environment is represented, but this is also true for the sky and time of day. The ability to activate and deactivate the smoke, or fire, was primarily thought to be reached through the hand-menu, and not to be included in the levels of the environment. This is to allow for a more controlled environment, and to avoid potentially overwhelming the patient by accidentally doing something unexpected. Furthermore, the ability to mute or unmute all audio present in the environment is also included in the functions of the hand-menu [19,21]. Also, there is a 'panic button', which if pressed by the therapist, changes the environment to the first level of exposure. Specifically, only the roads would be visible if the button was activated. The choice of having the roads as the lowest level of exposure was based on the logic of why a 'panic button' should not be controlled by the patient, but by the therapist [E1]. Patients

with PTSD have a tendency for avoidance, and therefore the connection to the environment, even if minimal, was viewed positively regarding the long-term therapeutic outcome [20,70].

7. EVALUATION OF CASE

In this chapter, the discussion and examination of the idea that the prototype was based on will be elaborated upon. This discussion and examination were with a psychiatrist who focuses on treatment of trauma and is familiar with most of the therapeutic techniques used.

The system evaluated was deemed beneficial and of high value, though suggestions were raised on improving specific aspects. One key improvement would be enabling the system to handle varying levels of realism while also offering depersonalization of objects within the environment. The introduction of relaxation and mindfulness modules was proposed. It's also crucial that therapists maintain control over the auditory components of the environment, which play a significant role in the therapy process. Another recommendation was incorporating tools such as an anxiety panel which visualizes the data from a PSS/SUDS scale, from 0 to 10. Regular measurement of the patient's experience, especially during exposure, would be useful. This scale could help assess whether the exposure is appropriately challenging or ineffective, as overly low scores might suggest the wrong type of exposure. This tool could also identify specific triggers, especially in cases where visual stimuli fail to provoke reactions, but sounds or other sensory input may. There was further discussion around incorporating physiological monitoring, such as heart rate, sweat detection, and SUDS – to determine which stimuli or methods work best. The system could also benefit from offering various general scenarios that therapists could customize further, and the inclusion of dynamic elements, like LLMs for interaction with figures, was suggested. Moreover, adding a functionality for the therapist to export detailed information from the system, including notes and measurements, could be valuable for assessing treatment progress. However, concerns around GDPR and data protection were raised, particularly in regard to the storage of sensitive information. The ability to shift between different roles in the environment, such as viewing the experience from another perspective, was also mentioned. Patients may benefit from the flexibility of navigating the environment in ways that provide a sense of control or distance, such as a helicopter view. Even symbolic transformations, like becoming a bird or an ant, could allow patients to approach trauma from a non-self perspective, offering a safer route to engage with difficult memories. Lastly, while additional stimuli such as smell and touch were mentioned, these were seen as secondary in importance. However, the concept of using powerful stimuli, like smell, could be valuable when traditional visual or auditory triggers lose their effect.

8. DISCUSSION

The aim of the thesis has been to explore the field of VR-based therapy for PTSD, and in doing so, attempting to answer the research questions guiding this work.

The use of VR systems in therapy for PTSD is not utilized to any higher degree judging from the interviewees unfamiliarity with the treatment [P1, P2, E1]. The literature, however, indicates a diligent exploration of the research and development in this field, with various systems being developed, such as BraveMind [20], and further research is continuously undertaken. The VR-based systems seem to be aimed increasingly towards a personalized, adaptable environment regarding the trauma and needs of the patient [16,24,39,42,48,73,74]. This approach is very important, offering a flexible and adaptable solution for the varied contexts of how the traumatic experience unfolds. Currently, VR-based therapy shows promise in several areas, improving immersion and patient engagement [25,44,57,70], but it may be beneficial to confirm its effectiveness across diverse PTSD cases [66]. The stimuli presented to the patient have primarily been visual and auditory, and indirect exposure, such as symbolic interaction, allow for a more safe and subtle approach [18,20,25,41,46]. The ability to steer away from realism as the goal, and instead explore alternative methods of exposure [18,41,46]. This could be vital in some cases and provides a more varied foundation [25,57], blending realism and symbolism. However, it seems that there is a need for an implementation that allows for real-time monitoring of the patient, to maintain the balance between efficacy and safety better [51,57,76]. This would allow for a higher degree of a tailored experience [51,57,76], as the ability to gauge reactions before they happen could be vital [E1]. Furthermore, the use of familiar stimuli, such as sounds or images, may aid in providing the patient with the feeling of being grounded in the present [25]. Indeed, the stimuli used is most commonly visual and auditory [22,38,57]. The inclusion of tactile, olfactory stimuli and movement may be one path forward [25,41,46,57]. This requires large improvements but can have a large impact on immersion and therefore engagement, also cases where the traumatic experience can be triggered more effectively via, for example, olfactory stimuli [22]. The ability for the patient to perceive, move, and react to stimuli may allow for a more intimate connection to the patient's trauma [38], and thus the therapeutic outcome [25,41,46,57]. The risk of isolation the patient may experience within the environment, may be mitigated by the co-presence of an avatar of the therapist [E1], but this may need further examination. The concept has been explored [25,44,55], and does seem relevant in the context of immersion. The addition of avatars for both patient and therapist, may allow for a more continuous treatment session [E1], and may be part of a patient safety protocol. More specifically, co-presence may provide a sense of guidance and safety for the patient but would also allow for a quicker intervention by the therapist if needed [E1]. Training tools for the therapists are needed to effectively utilize the tools if the situation arises [41,44]. It was also noted, that creating virtual environments from scratch is hard and complex, and may require a large investment of time, to achieve a good result. Therefore, a system like this should be developed as a framework with access to libraries of predeveloped objects and functions allowing for easy addition or adjustment of the environment, without the need for explicit programming. The therapist's ability to adjust the environment could offer a more effective, personalized experience for the patient

[62,66]. When new information of the trauma is gained, quick adjustments of triggers could be beneficial. The session could initially provide a generic environment [7,21], but as the sessions progress, it becomes more personalized to match what the patient shares.

A prototype was developed as proof of concept, allowing for the exploration of VR-based therapy for PTSD treatment. Two environments were designed: a winter forest scenario, and a city combat scenario. These environments were created based on fictional descriptions of traumatic experiences, aiming to simulate the conditions that would evoke PTSD responses in a controlled manner. The result of the thematic analysis (see appendix F) played a central role in informing the design of these environments. Objects and functionalities were included based on the therapeutic elements that emerged from the analysis [P1, E1,21,23,25,28,32,40,49,51,54,58,60,80]. However, due to hardware limitations the focus was primarily on visual and auditory stimuli. While the auditory aspects were only minimally explored, the implementation still allowed the therapist to disable or activate sound during the session [21,23,30,55]. The doctor also stressed the importance of giving therapists control over auditory components, as these play a crucial role in managing the patient's therapeutic process. The visual aspects were explored in much greater detail. One of the core functionalities implemented into the prototype was the ability to adjust the intensity of visual stimuli within the environment [30,55]. The controlled exposure to trauma-related stimuli is vital [19,54]. The prototype aimed to provide a flexible and adaptable therapeutic tool, for example, the option to switch between lower and higher quality objects within the environment. This was demonstrated through the military vehicles in the combat scenario. The lower-quality version of the vehicle was intentionally using green to denote a military object in low detail. The doctor proposed that it would be beneficial to add a feature that allows therapists to depersonalize objects within the environment, providing more control over how realism is introduced in the scenario. The importance of integrating dynamic visual elements that can be adjusted based on the patient's needs is a key factor for effective and manageable treatment [25]. Furthermore, the doctor suggested incorporating a relaxation or mindfulness module, which could be used in combination with therapy sessions to help patients manage stress during exposure. The question of patient safety was a red thread throughout the development of the prototype. The necessity of manageable controls of stimuli is vital to avoid overwhelming the patient, especially due to the perceived power VR has [19,60]. If the patient was overwhelmed, the therapist could press a panic button, which would revert the environment to the lowest possible stimuli level. The doctor also recommended the inclusion of a SUDS scale tool to regularly measure patient anxiety from 0 to 10. This tool could provide feedback on whether the exposure is challenging enough or too overwhelming, and help identify which sensory inputs, such as sound or visual stimuli, are most effective. Additionally, the doctor raised concerns about GDPR and data protection, particularly regarding the storage of sensitive patient data.

8.1. ETHICAL AND SOCIAL CONSEQUENCES

The ethical consequences of the utilization of VR-based therapy for PTSD are complex and possibly worrying. Research suggests that potential harm to patients during or after treatment may be under-reported [35]. The potential harm to patients, such as overstimulation, can be assessed through examining the literature on the impact of VR [19,60]. Indeed, exposure to trauma does inherently carry the risk of causing minor to major discomfort for the patient [69]. In cases where the patient doesn't remember the event, exposure may aid memory recall, creating a less controlled environment compared to when the therapist knows the event. Mitigating potential harm to patients is a common focus in the literature, and risk management will likely become very important as research progresses. However, the societal consequences of VR-based therapy are another area of interest. Everyone that has access to this type of treatment should, e.g., be protected through various safety implementations. But, the accessibility of this type of treatment being limited, due to cost and location, but also that this type of treatment is not as widespread as traditional ones. The treatment offered may not be continuously updated. Furthermore, the aspects of culture, the view on mental health and the wish to receive treatment may be frowned upon due to different social norms. The question of privacy and data security also becomes very important, as the environment and the monitoring may lead to highly sensitive data being stored. The risk of potential data breaches, or that the data would be misused, may lead to a strong mistrust of the technology. However, if these treatments become more accessible, attempts are continued to make it safe, and the societal view on it does not hinder the individual from requesting it, the positive impact it may have been large. There is even a possibility that individuals with treatment-resistant PTSD may be helped, where traditional therapy has lacked [25,26]. The risk of attempted suicide or substance abuse among these individuals due to untreated PTSD may become less common [17,27,29]. The possibility of improving the quality of life for many is alluring, but the path should be tread carefully, ensuring that it does not lead to unintended harm, despite being paved with good intentions.

8.2. FUTURE PERSPECTIVES

The aim of is to provide a safe but efficient treatment that is suitable for the varied needs of patients, regardless of the origins of their trauma. The ability to make it safe may necessitate the need for continuous monitoring of the patient's physical and psychological reactions [51,57,76]. With this data, it could be possible to create a system that classifies reactions to stimuli, leading to more accurate treatment. Accurate in the sense that the therapist could better predict potential reactions based on previous data. It may also allow for extrapolation to previously unused stimuli. However, the patient could be at risk of being overwhelmed before the system becomes fully accurate. One solution might be to expose the patient to various sample stimuli, based on the planned environment, which could then be extrapolated from. Additionally, anonymized third-party data could be combined with patient data to improve prediction accuracy, provided it matches the trauma origin. Importantly, the system should inform the therapist rather than making decisions autonomously to avoid false predictions. Nevertheless, monitoring could enhance both safety and the

precision of the treatment. The various stimuli that together shape the exposure the patient is subjected to may broaden, to more commonly include tactile, olfactory and movement. However, the question of if it would be necessary for the treatment to be effective may vary from patient to patient [1,63]. But the movement aspect is of interest, as it could facilitate a higher level of agency for the patient, and the improved ability of being able to interact with the environment [14]. This could allow for a variety of indirect exposure methods or allowing the empowerment of the patient to act differently, compared to the actual traumatic event [7,59]. It may be an asset to allow patients to explore different actions within a context, such as defending themselves or providing tools to de-dramatize the situation [38]. For example, provide the patient the ability to throw watermelons as a symbolic representation of an aggressor. Furthermore, it would be of interest to explore various roles within a traumatic context, such as viewing it from a bystander's point of view or the aggressors [6,41]. It may be more suitable in contexts where the perspective of the patient is distorted or, that guilt is a dominant feeling for the patient, as to attempt to reframe or broaden the perspective of the event [6,41]. Also, the possibility of integrating Augmented Reality functionality in these VR-systems could allow for a previously unprecedented type of exposure. It may be the closest one could come to mimicking reality and inputting a digital representation of triggering stimuli. Regardless of the various suggestions made above, the future of VR in a therapeutic setting seems very promising.

9. CONCLUSION

This thesis explored the potential of Virtual Reality as a therapeutic tool for treating PTSD. Through a combination of literature review and semi-structured interviews with both experts and individuals with PTSD, insights were gained regarding the current state of VR-based therapy and its application in mental health care. Additionally, a prototype was developed as part of this study, showcasing the feasibility of integrating VR into PTSD therapy. The findings suggest that VR therapy offers a unique opportunity to create immersive and controlled environments that can enhance traditional therapeutic methods. However, the research also highlighted challenges, such as safety, technological constraints and the ethical implications of using VR for trauma exposure. While the potential for VR therapy is clear, further research and development are necessary to address these challenges and to improve the application of technology for broader clinical use. Furthermore, the examination of Augmented Reality could be the next step in regard to the VR-systems ability for exposure. The limitations of this study, such as the small sample size, lack of user testing, and technical constraints, point toward future research possibilities. Further exploration is also needed to validate the effectiveness of VR therapy across different types of traumas and to ensure that the technology can be safely [66]. Beside these limitations, this thesis has contributed to the growing knowledge of VR therapy for PTSD, highlighting both the opportunities and the areas in need of examination. With continued research, development, and careful consideration of the ethical and practical challenges, VR has the potential to become a powerful tool in the treatment of PTSD and other mental health conditions.

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APPENDICES

Appendix A: Literature included within the Literature Review

Number	Title	Author	Year
1	Virtual Trauma Interventions for the Treatment of Post-traumatic Stress Disorders: A Scoping Review	[41]	2020
2	Feasibility of Virtual Reality Exposure Therapy in the Treatment of Danish Veterans with Post-Traumatic Stress Disorder: A Mixed Method Pilot Study	[20]	2023
3	Expanding Clinical Assessment for Traumatic Brain Injury and Comorbid Post-Traumatic Stress Disorder: A Retrospective Analysis of Virtual Environment Tasks in the Computer-Assisted Rehabilitation Environment	[61]	2017
4	Validating a Virtual Environment for Sexual Assault Victims	[46]	2017
5	An Innovative Framework for Delivering Psychotherapy to Patients With Treatment-Resistant Posttraumatic Stress Disorder: Rationale for Interactive Motion-Assisted Therapy	[25]	2018
6	Application of virtual reality technology in posttraumatic stress disorder	[77]	2023
7	The treatment and development prospect of VR exposure therapy for mental diseases	[76]	2022
8	Multisensory Virtual Reality Exposure Therapy	[51]	2018

9	A VR scene modelling platform for PTSD treatment	[70]	2017
10	A fully customizable truck-driving simulator for the care of people suffering from Post-Traumatic Stress Disorder	[18]	2016
11	Engineering a Showcase of Virtual Reality Exposure Therapy	[55]	2019
12	Virtual Reality Revolution: Strategies for treating mental and emotional disorders	[57]	2022
13	Application of Virtual Reality Technology in Psychotherapy	[44]	2020
14	Case report: the addition of olfaction to virtual reality enhanced exposure therapy for PTSD	[22]	2024
15	Users' Opinion About a Virtual Reality System as an ADjunct to Psychological Treatment for Stress-Related Disorders: A quantitative and qualitative mixed-methods study	[28]	2018
16	Applications and future perspectives of virtual reality in the treatments of post traumatic stress disorder	[48]	2021
17	A Literature Overview of Virtual Reality (VR) in Treatment of Psychiatric Disorders: Recent Advances and Limitations	[63]	2019
18	A Narrative Review of Virtual Reality Applications for the Treatment of Post-Traumatic Stress Disorder	[38]	2021
19	Virtual Reality in the Diagnostic and Therapy for Mental Disorders: A systematic review	[75]	2022
20	Tailored Immersion: Implementing Personalized Components into Virtual Reality for Veterans with Post-Traumatic Stress Disorder	[71]	2021

Appendix B: The inclusion criteria and Keywords used for the Literature Review

Table 1: Inclusion Criteria for the Literature Review

Inclusion Criteria	Description
1	The selected article should be written in English

2	The selected article should be peer reviewed
3	The selected article should be accessible
4	The selected article should be published between the year 2015 to 2024
5	The selected article should discuss the use of virtual environments in the context of therapy or treatment, preferably with a focus on PTSD

Table 2: Keywords, Databases, Results per Keyword foreach database

Keywords	PubMeds Search Results	IEEE Xplore Search Results	Uppsala University Library Search Tool
((("Virtual Reality" OR VR) AND (PTSD OR "Post traumatic stress disorder"))) AND ("Virtual Environment")) AND (Therapy OR Treatment)	9	48	96

Appendix C: All themes, the source and description.

Articles / Type of Interviewee	Theme	Description
[18,20,22,25,28,38,41,44,46,48,51,55,57,63,70,71,75–77]	Multifaceted Role of Virtual Environments in Therapy Context	The theme lifts the importance of utilizing personalized and symbolic VEs to enhance trauma therapy. Personalized VEs, tailored to the patients needs with visual, auditory and other stimuli, offer adjustable and flexible therapeutic experiences [18,22,38,41,44,51,77]. Symbolic interaction, using metaphors instead of realistic representations, provides less distressing engagement with traumatic memories [18,41,46]. It has been shown that a generic scenario, and not a replica of the trauma, is still effective in treatment [20]. Multi-sensory stimuli in VEs enhance immersion, decrease avoidance and boost emotional engagement, which is vital for therapy effectiveness [22,25,38,41,71]. VEs should invoke emotional responses, and if, depending on the needs of the patient, interaction and ability to alter the scenario, which can help patients with guilt related to their trauma [38,41]. It can be beneficial to create scenarios based on different emotions, as it can aid in evoking it within the patient [28]. While the practice via physical tasks within the VE is beneficial outside of it [76], patients with brain injuries may find it more challenging to benefit from it [75]. The balance between realism and symbolism is vital, and symbolic VEs offer a higher flexibility while maintaining effectiveness [25,57]. However, creating effective VEs is a hard and challenging process, and VR may also cause disorientation and discomfort for the patient [48,55,63,70].

[18,20,22,25,28,38,41,44,46,48,51,55,57,71,76]	The Impact and Essence of Immersion	The theme explores the important role of multisensory immersion and presence in VEs for therapeutic efficacy [18,22,25,38,41,71]. The importance of a sense of presence and realism in VEs, improved by multisensory stimuli, aid to the feeling of “being there” which relate to the effectiveness of the treatment [18,41,51,76]. The interactions within the VE can improve the sense of realism, which in turn can evoke emotional responses which are essential [46,57]. The sense of realism can be improved by including familiar scenes and stimuli [18], real-time adjustments [38,41], and multisensory experiences such as visual, auditory, olfactory and tactile stimuli [22,38,57]. The context and the situation that is visualized is strongly linked to immersion and not only sensory input [44]. When creating a VE, it is important to be aware and achieve spatial, motoric and cognitive immersion, as it improves the patient's ability to perform tasks [25,74]. The tasks can be beneficial, as it may allow for recreation of bodily experiences for the patient [20]. Realistic stimuli, familiar objects or environments, and the patient's engagement and agency within the VE, are important for realism and immersion [18,38]. However, the articles also suggest that VR may cause a sense of isolation for the patient, which can inhibit the efficacy of the treatment [48], but also improve it depending on needs of the patient [22]. It is also suggested that patients may benefit from a wristwatch or similar elements that provide them with information [55]. The usability and friendliness of the interaction within the VE for the patient is important, as it improves the overall experience [28].
[18,20,22,25,28,38,41,44,48,51,55,57,61,70,75,76]	The Optimization of Virtual Environments for Effective Treatment Outcomes	The theme focuses on creating controlled, repeatable and adjustable VEs to facilitate gradual exposure to traumatic stimuli. The role of the therapist is critical, as managing and adjusting the VE in real-time, ensuring that it aligns with the patient's therapeutic needs [28,55]. Integrating symbolic elements within the VE can aid in memory processing and increase the efficacy of the treatment [25]. Also, exploration of different roles in a traumatic setting may be beneficial [44], but also the agency in regard to reaction [38], self-defense for example, as it allows for a broader perspective. A combination of realistic and symbolic VEs may further improve memory retrieval and engagement, which improves the relation between therapist and patient [25]. Understanding patient performance and responses is essential [48], as VR therapy can trigger physical and psychological reactions [18,38,61,75]. Real-time psychological and physiological monitoring allows for more accurate adaptive therapy sessions [51,57,76]. It can also aid in measuring the patient's reaction to stimuli, which allow for more precise adjustment, but also to gauge the patient's progress over time [38]. For example, smells are effective for triggering PTSD symptoms, which highlights the need for careful control of sensory input [22]. By providing a safe and controlled environment, it is possible to mitigate avoidance behavior, which is essential for long-term symptom relief [20,70].
[20,22,25,28,38,41,51,57,70]	Auditory, Tactile, Visual and Olfactory Stimuli	The theme relates to a multisensory approach that focuses on the use of auditory, tactile, visual and olfactory stimuli to evoke trauma-related experiences. Auditory stimuli can be used in various stages, from warm-up and cooldown phases, to triggering emotional responses and recalling memories [20,25,28,38,41,51,57]. Tactile stimuli, such as vibrations and

		other physical sensations, enhance the sense of presence and immersion for the patient [20]. It improves the ability to re-enact specific experiences and can be vital for creating a realistic and engaging experience [20,38,41,51]. Visual stimuli are also vital, allowing one to view personal visual trauma stimuli in relation to dynamic visual elements related to specific environments, like a jungle or flying helicopters [25,38,41,51,57,70]. Olfactory stimuli contribute to a more holistic approach, triggering emotional memories and improving the realism of the scenes with smells like burning rubber, smoke, and gunpowder [20,22,38,51,57].
[25,38,41,48,51,57,63,77]	Movement	Physical movement towards visual triggers is highlighted as a positive action [20,57] that can lead to decreased avoidance [25,77] and improved cognitive processes [77]. However, the synchronization between avatar and patient is crucial for an immersive experience [51]. When this synchronization mismatches, it can result in disorientation, headaches, and eye fatigue for the patient [48,63]. Furthermore, the urge to act when exposed to stimuli, such as defending yourself or altering behavior, was noted [38]. This highlights the dynamic between psychological responses and physical movement [38].
Person 1 / P1	Adapting Therapeutic Approaches to Individual Needs	The theme focuses on the importance of adapting therapeutic approaches to meet individual needs, ensuring efficacy in addressing diverse challenges and achieving desired outcomes. The driving force for this theme was previous negative experiences with therapy, which inhibited and sometimes completely removed the efficacy of the treatment.
Person 1 / P1	Creating Safe Spaces in Therapy	The theme focuses on safety in therapy, particularly in the context of VR technology. To strive for increased safety and reduced risks, the introduction of safety protocols, and possible features within the virtual environment is crucial. Furthermore, experienced and well-trained personnel are necessary to minimize risks. The theme emerged from discussions of previous negative experiences, indicating the potential risk of therapy in general, translating into the context of VR-based therapy. The interviewee suggested bringing a support animal or object, to provide comfort during treatment.
Person 1 / P1	Patient-To-VR Specific Considerations	The theme highlights the importance of understanding a patient's perspectives and needs when using VR technology for therapy. It involves considering how aware patients are of the VE and how they perceive it. It also addresses how VR can benefit individuals with aphantasia, a condition where mental visualization is difficult or impossible. Another key aspect of this theme is ensuring a safe environment, as previous negative experiences can affect the perception and therefore trust.
Person 2 / P2	Challenges with Traditional Therapy	The theme represents the perceived challenges related to seeking and receiving traditional therapy that meet the needs of the patient. It involves aspects identified by the patient that influence the efficacy of the treatment, as well as desirable aspects of a more personalized therapy experience. The theme emerged via discussion of the difficulty finding effective treatment, and the importance of safety and experienced personnel.

Person 2 / P2	Perceptions of VR Therapy	The theme represents the mixed views on using VR in therapy. It involves aspects of fear, perceived risks, and the importance of mitigating these to strive for safety and control. This theme emerged from the individual's need for therapy while acknowledging the perceived risks of using such a powerful tool. However, the risks did not dissuade the interviewee's interest due to the potential benefits.
Person 2 / P2	Features of VR Therapy	The theme focuses on the features of VR therapy that provide a sense of control and comfort for the patient. It involves aspects related to control, safety, immersive experiences, and the ability to tailor the treatment to the patient's needs. This theme emerged from discussions highlighting the need for safety measures due to the immersive but potentially risky nature of VR therapy.
Expert 1 / E1	Innovative Therapy Through VR	This theme focuses on the transformative aspect of VR-based therapy, including the integration of traditional therapy techniques into a VR format and the adaptation to VR-specific approaches. It provides a perspective on the shift from regular therapy methods to immersive, technology-driven strategies, highlighting innovative possibilities and therapeutic benefits. The data that allowed this theme to emerge included discussions of various therapy techniques and their application in a virtual setting. It also included the necessity of training for personnel handling the treatment.
Expert 1 / E1	Patient Experience and Safety	This theme focuses on aspects regarding the patient's experience related to safety, comfort, and the techniques and tools taught within and outside the virtual environment. It involves considerations around patient-based safety measures and the challenge of balancing exposure with patient agency. The theme also emphasizes strategies to ensure comfort during therapy. The data from which this theme emerged highlighted the importance of patient safety and comfort due to risks associated with VR-based therapy. It also lifts the importance of therapeutic support and guidance during treatment, such as the co-existence of therapists and patients, to aid in-moment recovery from unforeseen reactions.
Expert 1 / E1	Personalization and Adaptive Therapy	The focus lies on the customization of the VR-based therapy to each individual patient's needs, such as the personalization of treatment plans, the adaptability of a VR environment, and the importance of patient-specific therapeutic content for efficacy and a safe level of patient engagement. This theme emerged from the discussions that the therapy needs to have a patient-focused approach. This is due to the unique differences in how individuals perceive and react, as well as their experienced trauma and its effects.
Expert 1 / E1	Communication and Emotional Processing in a Virtual Environment	This theme focuses on the unique challenges and benefits of emotional expression and therapist-patient communication within VR. It involves exploring how VR can facilitate new ways of expression and interaction, improving therapeutic processes. Additionally, the coexistence of both patient and therapist within the virtual environment is an aspect of this theme. The theme emerged via discussions on the innovative nature of VR-based therapy and the potential interactions therein that might broaden the ability for expression.

Expert 1 / E1	Navigating Ethical Terrain in VR Therapy	This theme's primary focus is on the ethical, safety, and research considerations when implementing VR-based therapy. It involves the responsible use of VR in therapy, the importance of evidence-based therapeutics, innovative practices, and continuous learning for refining and validating VR-based therapeutic approaches. This theme emerged from discussions on the perceived efficacy of VR and the potential risks it may bring.
Expert 1 / E1	Cognitive and Behavioral Change via VR	This theme focuses on using VR-based therapies to induce cognitive and behavioral changes. It involves techniques for restructuring, behavioral change, and building resilience through therapeutic approaches within VR. This theme emerged from discussions on various therapeutic techniques, their applications, and the adaptability of VR experiences.

Appendix D: Text from interviews, structured depending on what area and relevancy, which can inform the design of a virtual environment.

Nr	Source	Group	Relevancy for Virtual Environment	Data
1	P2	Therapy	Therapist co-existing in the VE.	".. I'm wearing a VR headset, part of my face is covered and they have to listen to me. Like if I say hey this is not going well, but if they are in the situation I am in. I feel like I could even maybe stay in the environment for longer because we can converse in there. "
2	P2	Tutorial	Tutorial that describes functionalities necessary for future therapy sessions	" I think that it could be good if, as in most VR games, there's a tutorial, something that shows you the ropes. Perhaps something like that could be good for VR too, because I assume that you can move or you can interact with the world in VR. "
3	P2	Patient	Symbol that indicates when session is active or not	" I would like to have some type of signal when a session is on and when a session is off. "

4	P2	Patient	Symbol representing function for stopping the session?	".. and I also feel that in me that it is a safe environment, then that's also vital. And perhaps...what if there was a red stop button? So I too can stop. "
5	P1	Patient	Digital representation which allows for recovery from traumatic stimuli?	"I usually bring a support animal , an animal which can be used to calm me down. "
6	P1	Patient	Function for exiting the VR environment?	"There is a worry of you reliving a memory in the VR space... I worry about the feeling of being trapped. "
7	P1	Therapist	Therapist can control the VE?	"..she will stop what is going on, and guide me back to feeling okay, and make sure that I am okay. "
8	E1	Therapy	Gradually increasing exposure	"The therapist has to really think and make the VR so that it's not too stressful for the client, expose client to something that's manageable in the beginning"
9	E1	Therapist	Ability to adapt the exposure	"It would be quite nice to be able to change the VR while the patient is having the exposure "
10	E1	Therapy	Co-existence within the VE.	"It would be great to be able to co-exist in a VR environment...you're there with that patient and you're guiding them."
11	E1	Therapy	Gradually Increasing Exposure	"You have to talk to the patient and understand where the threshold is, what they can actually accomplish..."

12	E1	Patient	Specific exposure based on patient trauma	“these things obviously are going to trigger this person also, it's not just that she was robbed by some men with the black dog.”
13	E1	Sound	Multi-sensory approach for exposure	“That could be a sound. It could be the trees. In the wind, it could be anything “
14	E1	Olfactory	Use of smell	“So if the client is triggered, then by the smoke for example”
15	E1	Therapist	Control of the VE, gradual exposure	“And then, but it would be nice if the client then could say OK, you can add the black dog now. And if the client could then say OK, now I'm ready to see like one of the men. “
16	E1	Therapy	Environment as a starting point?	“We came to an agreement that we would just expose it to the forest.”
17	E1	Therapy	Red button reinforces avoidance behavior	“It can be terribly destructive if you press the red button whenever the client is scared”

Appendix E: The two text descriptions that describe the two scenarios.

Text 1	Text 2
<i>A scenario takes place in an environment where the feeling of being isolated is discernible, as the snow-covered landscape is broken by a small forest. As the patient approaches the forest, and the cabin that is located deeper, a couple of men come into view. Their car is parked adjacent to the cabin, and they initially do not seem hostile. However, their demeanor shifts as they notice the patient. They have a dog on a leash, and it begins to bark, as two of the men approach, making suggestive and inappropriate comments.</i>	<i>A combat-inspired scenario takes place in a city environment. The streets are crowded with pedestrians and lined with vehicles waiting at stoplights. It's late in the day, as indicated by the fading sunlight. Suddenly, an explosion occurs at the front of the vehicle line, and a humvee bursts into smoke and flames. Gunfire erupts, causing panic as people scatter in all directions. The patient leaves their humvee, attempts to get a sense of control by positioning themselves behind the vehicle. This is when they are dangerously close to being hit from the armed assailants who open fire down the road in their direction.</i>

Appendix F: Virtual Environment Requirements Based on Gathered Literature and Interviews

Function/Elements/Aspects - Description	Reference
Include a warm-up, cooldown?	[24,38]
Include a tutorial that guides patients through functionality?	[P2, [39]
Patients starting-point is neutral	[E1]
Interactive elements where patients identify and challenge negative thoughts	[44,54]
Modules that teach relaxation techniques	[E1, 26,44]
Educative module, educate about PTSD, symptoms and recovery process	[18,41,77]
Facilitate expression	[E1]
Patient should have an avatar that represents users' limbs	[46,48,56]
Symbol to indicate if session is active or not	[51]
Environment should have sounds	[18,35]
Patient should be able to interact with elements	[44,55]
Patient should have agency within the virtual environment	[35]
Patient should be able to explore different roles, including self-defense scenarios	[42]
The users experience within the VE should be user-friendly	[27]

Patient can bring in a support object?	[P1]
Therapist should have ability to alter the virtual environment in real-time	[27,53]
Therapist should have an interface, to allow adjustment and managing the VE	[27,53]
Therapist should be visible in the VE	[E1]
Therapist should have safety protocols for patients safety	[P1, E1]
Therapist should be able to stop the session	[P1, E1]
Therapist should be able to decrease and increase the stimuli shown to the patient	[49,55,73]
Therapist should be able to decide what can be heard	[20,22]
System to classify and rank patient reactions to exposure	[35]
Audio to indicate START and END of session	[26,40]
Symbolic Stimuli such as images, objects related to trauma	[25]
Customizable music and sounds	[20,22]
Sounds/music that match the time of trauma, but also use present-time music to showcase end of session	[20,22,35,49,55]

Appendix G: The requirements that were able to be implemented in the two virtual scenarios which were created for this thesis.

Winter Forest Environment	City Combat Environment	Description

Yes	Yes	Initial Role Request (Patient or Therapist)
Yes	Yes	Both Patients and Therapists have - Visible Hands - Ability to move - Ability to grab things
Yes	Yes	Patients have neutral starting position
Yes	Yes	Therapists have - Ability to alter time of Day - Ability to alter weather
Yes	Yes, but to a higher degree	- Ability to minimize and increase environmental stimuli
Yes	Yes	- Panic Function (Lowest stimuli possible)
Yes	Yes	- Ability to mute/unmute audio
Yes	Yes	- Ability to activate special effects (Snow, Smoke, Fire)
Yes	Yes	- A hand menu, showcasing all the abilities/functions therapists have
Yes	Yes	- Figures that can be controlled by therapist
Yes	Yes, but to a higher degree	- Figures that offer multiple levels of stimuli